

ETF Proxy Models

NAV, Risk and Hedging

Goldman Sachs Quant Hackathon 2026 — Problem Statement

Training Data: Full Year 2025

1. Background

1.1 What is an ETF?

An Exchange Traded Fund (ETF) holds a basket of underlying securities and trades on an exchange like a stock. Its Net Asset Value (NAV) represents the fair value of its holdings per share. Unlike mutual funds, ETFs trade continuously and their market price can deviate from NAV — creating opportunities and risks for market participants.

1.2 ETF Market Making

Market makers continuously quote bid/ask prices, providing liquidity to buyers and sellers. The core activities are:

- Pricing — estimating ETF fair value (NAV) in real time
- Quoting — setting bid/ask prices around fair value, capturing the spread as revenue
- Inventory management — tracking net positions when buy/sell flow is asymmetric
- Hedging — offsetting market risk of accumulated inventory using liquid instruments

1.3 Fixed Income ETFs — Why They Are Hard

Unlike equity ETFs whose underlyings are exchange-traded and continuously observable, fixed income ETFs present unique challenges:

- Complex compositions — a single ETF may hold hundreds of bonds spanning IG corporates, high yield, government bonds, MBS, CLOs and more
- Illiquid underlyings — bonds trade OTC, often infrequently, with no continuously observable price
- Multi-dimensional risk — simultaneous exposure to interest rate risk across the curve and credit spread risk across sectors
- No central pricing — bond prices are often stale, estimated, or matrix-priced

1.4 The Core Problem

For equity ETFs, NAV estimation is straightforward — sum live prices of underlyings. For fixed income ETFs, full look-through is often not feasible. OTC instruments lack reliable real-time prices, and pricing hundreds of bonds individually is computationally intensive.

This creates two practical problems for a market maker:

- What is the NAV? — a proxy-based estimate of fair value is needed without full look-through
- What is the risk? — aggregate rate and credit sensitivity of ETF inventory is needed to hedge efficiently using liquid instruments such as Treasury futures and credit indices

This problem asks you to solve both challenges and then apply your solutions to manage a real portfolio.

2. A Note on Modelling Approach

Any model for a fixed income ETF can theoretically be expressed as a weighted combination of the same underlying market instruments. However, the choice of proxy instruments matters. Different sets of proxy instruments capture different components of ETF returns — some capture broad market movements, others isolate specific risk dimensions, and some are better at replicating total price behaviour.

A model that works well for one purpose may not be optimal for another. Candidates are encouraged to think carefully about what each part of the problem is asking for, and whether the same set of proxy instruments serves both objectives equally well.

Candidates are free to choose any modelling technique and any combination of the allowed proxy instruments. The problem does not prescribe a specific method. You will be evaluated purely on the accuracy of your outputs.

3. Universe of Instruments

3.1 Target ETFs

The following 5 ETFs are the subject of this problem. Their identities have been masked.

ETF
Target ETF_1
Target ETF_2
Target ETF_3
Target ETF_4
Target ETF_5

3.2 Proxy Instruments

The following instruments are provided for use in your models. You may use any combination of them, subject to the restrictions stated in each part of the problem.

3.2.1 Proxy ETFs

ETF
Proxy_ETF_1
Proxy_ETF_2
Proxy_ETF_3
Proxy_ETF_4
Proxy_ETF_5
Proxy_ETF_6
Proxy_ETF_7
Proxy_ETF_8
Proxy_ETF_9
Proxy_ETF_10

3.2.2 On-the-Run Treasury Bonds

Instrument
UST_2Y
UST_5Y
UST_10Y
UST_30Y

3.2.3 Treasury Futures

Instrument	Tenor
TU	2Y
FV	5Y
TY	10Y
US	20–25Y

3.2.4 CDX

Instrument
CDX_IG_5Y
CDX_IG_10Y
CDX_HY_5Y

CDX_HY_10Y

4. Data Provided

4.1 EOD_prices.csv

Daily time series of end-of-day price marks over the year 2025. First column = date. Remaining columns:

Columns	Description
Target_ETF_1 ... Target_ETF_5	Published EOD NAV per share for each target ETF
Proxy_ETF_1 ... Proxy_ETF_10	Published EOD NAV per share for each proxy ETF
TU, FV, TY, US	EOD prices per 100 par notional for Treasury futures
UST_2Y, UST_5Y, UST_10Y, UST_30Y	EOD prices per 100 par notional for on-the-run Treasury bonds
CDX_IG_5Y, CDX_IG_10Y, CDX_HY_5Y, CDX_HY_10Y	EOD prices per 100 par notional for CDX instruments

4.2 Return Definition

All proxy models must be expressed in terms of simple daily returns of the underlying instruments. Use the following definition throughout:

$$r_t = (P_t - P_{t-1}) / P_{t-1}$$

Do not use log returns or price levels.

4.3 Training and Test Data

- Training data: Full year 2025 — provided to candidates for model development and validation
- Test data: Hidden dataset held by organisers — used to evaluate submitted solutions
- Candidates should ensure their models generalise beyond the training period

5. Test Case Structure

Each test case is provided as a JSON object with the following structure:

```
{
  "mode":      "validate" or "evaluate",
  "key":       "nav" or "risk" or "hedge",
  "portfolio": { "Target_ETF_1": notional, ... } // hedge only
}
```

5.1 Two Modes

validate

Checks whether the candidate's submitted dataframe for the given key has the correct schema — row labels, column names, data types. All validate test cases are public and visible to candidates. Their purpose is to let candidates confirm their output format is correct before final submission.

evaluate

Actually scores the candidate's submission for the given key against the hidden test dataset. Most evaluate test cases are hidden from candidates, with a small number exposed as samples to demonstrate how scoring works.

5.2 Static vs Input-Dependent Submissions

An important property of this problem is that the NAV and Risk model submissions are STATIC — they are built once from the 2025 training data and do not change across test cases. The test input does not affect these submissions in any way.

The Hedge submission, however, IS input-dependent. The starting portfolio to hedge comes from the portfolio key in the test input JSON. Candidates must extract this portfolio from the input and construct a static hedging basket specific to that portfolio.

Submission	Depends on Test Input?	Notes
NAV model	No	Static dataframe trained on 2025 reference data, same across all test cases
Risk model	No	Static dataframe trained on 2025 reference data, same across all test cases
Hedge basket	Yes	Constructed per test case based on the portfolio key in the input

6. Part 1: NAV Estimation (1/3 of total score)

6.1 Problem

Build a proxy model for each Target ETF to estimate its end-of-day NAV. Each Target ETF's daily return must be expressed as a weighted combination of proxy instrument daily returns.

For Part 1 you may use any combination of the following proxy instruments:

$$\text{ALL_PROXY} = \text{Proxy_ETFs} + \text{TSY_bonds} + \text{TSY_futures} + \text{CDX}$$

6.2 Submission Format

Submit one dataframe:

- One row per Target ETF
- One column per proxy instrument from ALL_PROXY
- Cell (i, j) = weight of proxy instrument j in the NAV model of Target ETF i
- Set to zero if that instrument is not used

Placeholder example:

ETF	Proxy _ETF_ 1	Proxy _ETF_ 2	...	TU	TY	...	UST_ 2Y	...	CDX_I G_5Y	...
Target_ET F_1	0.60	0.40	...	0.00	0.00	...	0.00	...	0.00	...
Target_ET F_2	0.00	0.30	...	0.20	0.00	...	0.00	...	0.30	...
Target_ET F_3	0.00	0.00	...	0.00	0.00	...	0.50	...	0.00	...
Target_ET F_4	0.00	0.00	...	0.30	0.70	...	0.00	...	0.00	...
Target_ET F_5	0.20	0.20	...	0.00	0.00	...	0.00	...	0.40	...

6.3 Evaluation

Your NAV model is applied to the hidden test period prices to reconstruct predicted NAVs. We then compute the Mean Absolute Percentage Error (MAPE) between predicted and actual published NAV, per Target ETF:

$$\text{MAPE}_i = \text{mean}(|\text{predicted_NAV}_{i,t} - \text{actual_NAV}_{i,t}| / \text{actual_NAV}_{i,t})$$

The overall NAV score is based on the mean of MAPE values across all 5 Target ETFs. Lower MAPE gives a higher score.

7. Part 2: Risk Estimation (1/3 of total score)

7.1 Problem

Build a proxy model for each Target ETF to estimate its rate and credit risk sensitivities. As with Part 1, each Target ETF's daily return must be expressed as a weighted combination of proxy instrument daily returns.

For Part 2 you may only use non-ETF proxy instruments:

$$\text{RISK_PROXY} = \text{TSY_bonds} + \text{TSY_futures} + \text{CDX}$$

Proxy ETFs are NOT allowed in the risk model. The reason is that proxy ETFs mix rate and credit exposures, making clean attribution to specific risk dimensions impossible.

7.2 Submission Format

Submit one dataframe:

- One row per Target ETF
- One column per non-ETF proxy instrument (TSY_bonds + TSY_futures + CDX)
- Cell (i, j) = weight of proxy instrument j in the risk model of Target ETF i
- Set to zero if that instrument is not used

Placeholder example:

ETF	TU	FV	TY	US	UST_2Y	UST_5Y	UST_10Y	UST_30Y	CDX_IG_5Y	CDX_IG_10Y	CDX_HY_5Y	CDX_HY_10Y
Target_ETF_1	0.10	0.00	0.40	0.00	0.00	0.00	0.00	0.00	0.50	0.00	0.00	0.00
Target_ETF_2	0.05	0.00	0.50	0.00	0.00	0.00	0.00	0.00	0.40	0.10	0.00	0.00
Target_ETF_3	0.00	0.00	0.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.60	0.00
Target_ETF_4	0.10	0.00	0.30	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Target_ETF_5	0.10	0.00	0.20	0.00	0.00	0.00	0.00	0.00	0.20	0.00	0.00	0.00

Note: The conversion of your model weights into actual DV01 and CS01 values is handled by the evaluators. Candidates do not need to perform this conversion themselves.

7.3 Evaluation

Your model weights are converted by the evaluators into bucketed DV01 (buckets: 2Y, 5Y, 10Y, 30Y) and total CS01 for each Target ETF. These are compared against an internal benchmark held by the organisers.

Per-ETF errors are computed using L2 norm:

$$\begin{aligned} \text{L2_DV01_i} &= \sqrt{\sum \text{over } k \text{ in } \{2Y, 5Y, 10Y, 30Y\} \text{ of } (\text{DV01_i,k_model} - \text{DV01_i,k_bench})^2} \\ \text{L2_CS01_i} &= | \text{CS01_i_model} - \text{CS01_i_bench} | \end{aligned}$$

The overall risk score is a weighted sum of L2_DV01 and L2_CS01 contributions across all 5 Target ETFs. Lower L2 errors give a higher score.

8. Part 3: Portfolio Hedging (1/3 of total score)

8.1 Problem

You are given a fixed portfolio of long and short positions across Target ETFs in the test input (under the portfolio key). Using any proxy instruments (including ETFs), construct a static hedging basket that minimises the variance of the combined portfolio PnL and maximises hedge effectiveness over the test period.

All notionals throughout this section, both in the starting portfolio and in your hedging basket submission, are in mm USD (millions of US dollars). A value of 100 means \$100 million notional.

8.2 Starting Portfolio

The starting portfolio is provided in the test input JSON under the portfolio key, structured as a mapping from Target ETF names to signed notionals (positive = long, negative = short). For example:

ETF	Notional
Target ETF_1	150
Target ETF_2	-120
Target ETF_3	180
Target ETF_4	-100
Target ETF_5	130

8.3 Hedging Instruments

You may use any combination of the following instruments for hedging:

- Proxy ETF_1 through Proxy ETF_10
- Treasury futures: TU, FV, TY, US
- On-the-run Treasury bonds: UST_2Y, UST_5Y, UST_10Y, UST_30Y
- CDX: CDX_IG_5Y, CDX_IG_10Y, CDX_HY_5Y, CDX_HY_10Y

You may NOT use the Target ETFs themselves as hedging instruments.

Hint: You may use either your NAV proxy model from Part 1 (replication-style hedging) or your risk proxy model from Part 2 (risk-based hedging) to inform your choice of hedging instruments. Each approach has different strengths.

8.4 Transaction Costs

Setting up the hedging basket incurs transaction costs at inception, based on the notional traded per instrument:

Instrument	Cost
Treasury futures (TU, FV, TY, US)	0.5bp of notional traded
On-the-run Treasury bonds	0.5bp of notional traded
CDX instruments	1bp of notional traded
Proxy ETFs	2bp of notional traded

8.5 Submission Format

Submit one dataframe representing your static hedging basket:

- Index = names of proxy instruments you choose to use
- Single column labeled Notional
- Values = signed notional position in mm USD (positive = long, negative = short)

Placeholder example:

Instrument	Notional
TU	-10
TY	-220
US	185
CDX_IG_5Y	-200
CDX_HY_5Y	-155

8.6 Evaluation

The hedge is applied at day 0 and held static through the hidden test period. The evaluation is a weighted combination of three metrics:

Hedge Effectiveness Ratio (HER)

$$\begin{aligned}\text{Portfolio_PnL}(t) &= \text{sum over ETFs of } (\text{notional}_i \times \text{ETF_return}_i, t) \\ \text{Hedge_PnL}(t) &= \text{sum over hedge instruments of } (\text{notional}_j \times \text{instrument_return}_j, t) \\ \text{Net_PnL}(t) &= \text{Portfolio_PnL}(t) + \text{Hedge_PnL}(t)\end{aligned}$$

$$\text{HER} = 1 - \text{Var}(\text{Net_PnL}) / \text{Var}(\text{Portfolio_PnL})$$

Higher HER is better. HER = 1 means the hedge perfectly offsets all variance.

Tracking Error

$$\text{Tracking_Error} = \text{StdDev}(\text{Net_PnL}) / \text{StdDev}(\text{Portfolio_PnL})$$

Lower tracking error is better. Tracking error close to 0 means the hedged portfolio has near-zero daily PnL volatility.

Transaction Cost

Total transaction cost incurred when setting up the hedging basket. Lower is better. Penalises using too many or unnecessarily large hedge positions.

The overall hedging score is a weighted combination of these three metrics. Solutions with higher HER, lower tracking error and lower transaction cost score better.

9. Overall Scoring

Part	Focus	Weight
Part 1	NAV Estimation	1/3
Part 2	Risk Estimation	1/3
Part 3	Portfolio Hedging	1/3

Good luck.